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Safety data sheet

BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.
Date / Revised: 03.08.2024 Version: 1.0
Date / Previous version: not applicable Previous version: none
Product: **Ammonium carbonate Food Grade (E503i)**
(ID no. 30042216/SDS_GEN_IE/EN)
Date of print 10.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Ammonium carbonate Food Grade (E503i)

UFI: 6QCP-10MC-N006-VXEY

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: food additive(s)
Recommended use: food additive(s), Raw material

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF Ireland DAC
Asgard House, 19-20 City Quay
Dublin, D02 K744
Ireland

Telephone: +353 21 451-7100
E-mail address: product-safety-uk-and-ireland@basf.com

1.4. Emergency telephone number

For products classified as hazardous in accordance with CLP:
National Poisons Information Centre, Beaumont Hospital, Dublin 9
Emergency medical information: 8am-10pm (seven days)
Tel.: 01 8092566
International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

Acute Tox. 4 (oral)	H302 Harmful if swallowed.
Eye Dam./Irrit. 2	H319 Causes serious eye irritation.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

H319	Causes serious eye irritation.
H302	Harmful if swallowed.

Precautionary Statements (Prevention):

P280	Wear eye protection.
P270	Do not eat, drink or smoke when using this product.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P330	Rinse mouth.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Hazard determining component(s) for labelling: Ammonium carbamate, Ammonium hydrogencarbonate

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
Irritating to eyes, respiratory system and skin (dust).

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria. Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

Preparation based on:Ammonium carbamate, Ammonium hydrogencarbonateH_2CO_3 . x NH_3
CAS: 10361-29-2 EINECS: 233-786-0

Regulatory relevant ingredients

Ammonium carbamate

Content (W/W): 50 % CAS Number: 1111-78-0 EC-Number: 214-185-2 REACH registration number: 01-2119493982-22	Acute Tox. 4 (oral) Eye Dam./Irrit. 1 H318, H302

Ammonium hydrogencarbonate

Content (W/W): 50 % CAS Number: 1066-33-7 EC-Number: 213-911-5 REACH registration number: 01-2119486970-26	Acute Tox. 4 (oral) H302

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

If inhaled:
After inhalation of decomposition products: Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Overexposure may cause:, vomiting, dyspnea, nausea, coughing

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: After inhalation of decomposition products: Pulmonary odema prophylaxis. Treat according to symptoms (decontamination, vital functions), no known specific antidote, administer corticosteroid dose aerosol to prevent pulmonary odema.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, foam

5.2. Special hazards arising from the substance or mixture

Endangering substances: ammonia, anhydrous, Carbon dioxide
Advice: The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Further information:
Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Breathing protection required. Ensure suitable air extract/ventilation during cleaning/emptying of process machinery.

6.2. Environmental precautions

Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For residues: Sweep/shovel up.
Avoid raising dust.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Breathing must be protected when large quantities are decanted without local exhaust ventilation. Processing machines must be fitted with local exhaust ventilation. Avoid dust formation.

Protection against fire and explosion:
Store in a cool place. If heated the drums can burst due to pressure build-up.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from nitrites and alkaline substances. Storage and transport only combined with food materials or food additives. Separate from flavoring agents.
Do not store with: Sodium nitrate, sodium nitrite

Suitable materials for containers: Aluminium, High density polyethylene (HDPE), glass, Low density polyethylene (LDPE), Stainless steel 1.4541, Stainless steel 1.4571, enamelled, rubberized
Further information on storage conditions: Keep container in a well-ventilated place. Keep container dry.

Storage stability:
Storage temperature: < 30 °C
The stated storage temperature should be noted.

Protect from temperatures above:30 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

No substance specific occupational exposure limits known.

Components with PNEC

1111-78-0: Ammonium carbamate
freshwater: 0.418 mg/l
marine water: 0.0418 mg/l
intermittent release: 0.37 mg/l

sediment (freshwater): 1.89 mg/kg
sediment (marine water): 0.189 mg/kg
soil: 0.133 mg/kg
STP: 10 mg/l

1066-33-7: Ammonium hydrogencarbonate
freshwater: 0.37 mg/l
marine water: 0.037 mg/l
intermittent release: 0.63 mg/l
sediment (freshwater): 0.1332 mg/kg
sediment (marine water): 0.01332 mg/kg
soil: 74.9 mg/kg
STP: 1347 mg/l

Components with DNEL

1111-78-0: Ammonium carbamate
worker: Long-term exposure- systemic effects, dermal: 14.1 mg/kg
worker: Long-term exposure- systemic effects, Inhalation: 49.8 mg/m3
consumer: Long-term exposure- systemic effects, dermal: 7.1 mg/kg
consumer: Long-term exposure- systemic effects, Inhalation: 12.3 mg/m3

1066-33-7: Ammonium hydrogencarbonate
worker: Long-term exposure - systemic and local effects, Inhalation: 62.5 mg/m3
worker: Long-term exposure- systemic effects, dermal: 57 mg/kg
worker: Short-term exposure - systemic and local effects, Inhalation: 160.7 mg/m3
consumer: Long-term exposure - systemic and local effects, Inhalation: 13.33 mg/m3
consumer: Short-term exposure - systemic and local effects, Inhalation: 143.91 mg/m3
consumer: Long-term exposure- systemic effects, dermal: 34.2 mg/kg

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1) Suitable respiratory protection for higher concentrations or long-term effect: Self-contained breathing apparatus.

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
polyvinylchloride (PVC) - 0.7 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Do not breathe dust. At the end of the shift the skin should be cleaned and skin-care agents applied.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	solid	
Form:	crystalline, powder	
Colour:	white	
Odour:	strong, ammonia-like	
Odour threshold:	No data available., not determined	
Melting point:	not applicable The substance / product decomposes.	
Boiling range:	Study technically not feasible., The substance / product decomposes therefore not determined.	
Flammability:	not highly flammable, not flammable, not self-igniting	(Regulation 440/2008/EC, A.10)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Flash point:	not applicable, the product is a solid	
Auto-ignition temperature:	The substance / product decomposes therefore not determined.	
Thermal decomposition:	> 59 °C (internal method) To avoid thermal decomposition, do not overheat.	
pH value:	9 (100 g/l, 20 °C)	(pH Meter)
Viscosity, kinematic:	not applicable, the product is a solid	

Viscosity, dynamic:	not applicable, the product is a solid	
Solubility in water:	320 g/l (20 °C)	(internal method)
Partitioning coefficient n-octanol/water (log Kow):	-2.4 - -0.47	
Vapour pressure:	69 mbar (20 °C) Literature data. 188 mbar (30 °C) Literature data.	
Density:	approx. 1.6 g/cm3 (20 °C)	(OECD Guideline 109)

Particle characteristics

Particle size distribution:	300 - 400 µm	(D50, measured)

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating.

Other safety characteristics

Bulk density:	780 - 830 kg/m3	(other)
Evaporation rate:	negligible, The product is a non-volatile solid.	

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Exothermic reaction. Reacts with nitrites. Reacts with nitrates.

10.4. Conditions to avoid

Avoid heat. See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:
strong bases

10.6. Hazardous decomposition products

Hazardous decomposition products:
ammonia, anhydrous, Carbon dioxide

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion.

Experimental/calculated data:

LD50 rat (oral): > 1,800 - < 2,150 mg/kg (BASF-Test)
LD50 rat (dermal): > 2,000 mg/kg
No mortality was observed.

Information on: Ammonium hydrogencarbonate
Experimental/calculated data:
LC50 rat (by inhalation): > 4.74 mg/l 4.5 h (other)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. An aerosol was tested.

Information on: Ammonium carbamate
Experimental/calculated data:
LC50 rat (by inhalation): 6.6 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The vapour was tested.

Irritation

Assessment of irritating effects:

Not irritating to the skin. Eye contact causes irritation.

Experimental/calculated data:

Serious eye damage/irritation

rabbit: Irritant. (OECD Guideline 405)

Information on: Ammonium carbamate
Assessment of irritating effects:
May cause severe damage to the eyes. Not irritating to the skin.

Information on: Ammonium hydrogencarbonate
Assessment of irritating effects:
Not irritating to the eyes. Not irritating to the skin. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Respiratory/Skin sensitization

Assessment of sensitization:

The chemical structure does not suggest a sensitizing effect.

Germ cell mutagenicity

Assessment of mutagenicity:

No data available concerning mutagenic effects. The chemical structure does not suggest a specific alert for such an effect.

Information on: Ammonium hydrogencarbonate
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture.

Information on: Ammonium carbamate
Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Information on: Ammonium hydrogencarbonate
Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: Ammonium carbamate
Assessment of carcinogenicity:
Did not show carcinogenic effects in animal experiments. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Information on: Ammonium hydrogencarbonate
Assessment of reproduction toxicity:
Study scientifically not justified.

Information on: Ammonium carbamate
Assessment of reproduction toxicity:
Study scientifically not justified.

Specific target organ toxicity (single exposure)

No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

No data available.

Aspiration hazard

No data available.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms.
Acutely harmful for aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The product has not been tested. The statement has been derived from the properties of the individual components.

Toxicity to fish:
LC50 (96 h) 61 mg/l, Oncorhynchus mykiss (Flow through.)
Literature data.

Aquatic invertebrates:
EC50 (48 h) 63.7 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
Nominal concentration.

Aquatic plants:
EC50 (72 h) 75.9 mg/l (biomass), Desmodesmus subspicatus (DIN 38412 Part 9, static)

Microorganisms/Effect on activated sludge:
EC20 (0.5 h) 1,000 mg/l, activated sludge, domestic, non-adapted (OECD Guideline 209, aquatic)

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Inorganic product which cannot be eliminated from water by biological purification processes. Can be oxidized to nitrate, or be reduced to nitrogen, by microorganisms.

12.3. Bioaccumulative potential

Bioaccumulation potential:
Accumulation in organisms is not to be expected.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria.

The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Test for use in agriculture.

SECTION 14: Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

Further information
Specific national features of transport regulations must be observed. They are to be found in the shipping documents.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

15.2. Chemical Safety Assessment

Chemical Safety Assessment not required

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Eye Dam./Irrit. 2A
Acute Tox. 4 (oral)
Aquatic Acute 3

Any other intended applications should be discussed with the manufacturer.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
H319	Causes serious eye irritation.
H302	Harmful if swallowed.
H318	Causes serious eye damage.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational

Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.